8.1 Parasite and Host

Parasites are living organisms, which depend on living host for their nourishment and survival. They multiply or undergo development in the host. Host is defined as an organism, which harbors the parasite, provides nourishment and gives shelter to parasite. Host is relatively larger than the parasite.

8.1.1 Association between Host and Parasite

The relationship between host and the parasite can be of the following types:

- Symbiosis
- Commensalism, and
- Parasitism.

Flowchart 8.1 describes the types of host – parasite relationship

8.1.2 Types and Classification of Parasite

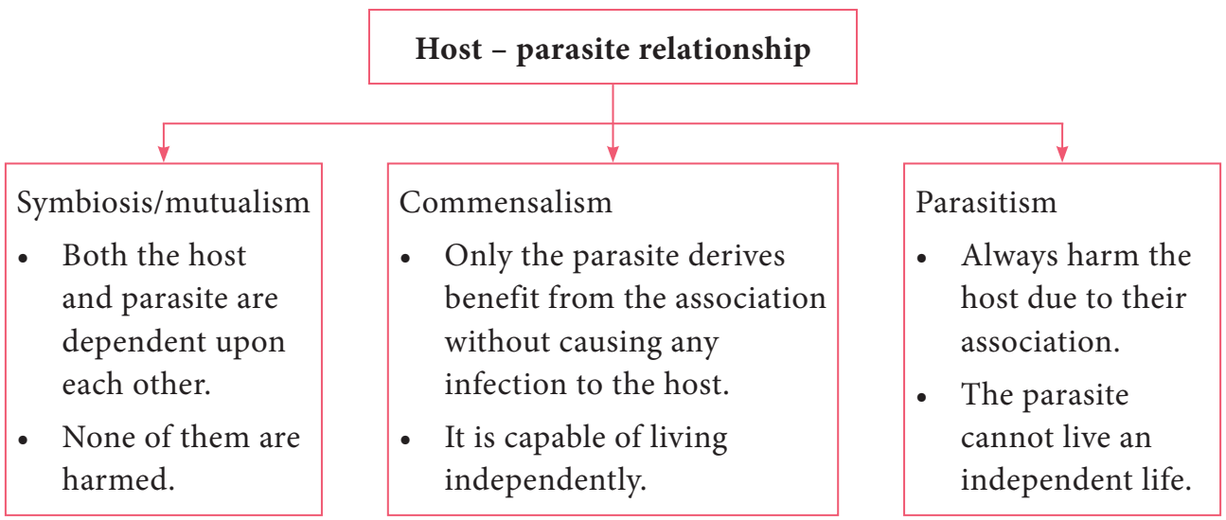
According to the nature of the host- parasite interaction and the environmental factors, the parasite may be one of the following,

Ectoparasite: These parasites live on the outer surface or in the superficial tissues of the host (Example: Lice). The infection by these parasites is called **infestation**.

Endoparasite: The parasite which lives within the host is called Endoparasite. Invasion by the parasite is called **Infection**. Most of the protozoan and helminthic parasites causing human diseases are endoparasites.

Endoparasites can be further classified as:

- **Obligate parasite:** This parasite is completely dependent on its host and cannot survive without it. Example: *Hookworms*.
- **Facultative parasite:** This parasite may either live as free living form or as a parasite when the opportunity arises. Example: *Naegleria fowleri*.
- **Opportunistic parasite:** This parasite is capable of producing disease in an immune deficient host (like AIDS and cancer patients). Example: *Toxoplasma gondii*.
- **Zoonotic Parasite:** This parasite primarily infects animals and is



Flowchart 8.1: The types of host – parasite relationship

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